

MATH1231 CALCULUS

Chapter 3A

Term II 2026

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June 1, 2026

3. Ordinary Differential Equations

(3.1) Introduction

Engineering and Science are built on models of physical reality.

Mostly, these models are mathematical models which allow predictions, analysis, and discovery of new relationships.

Many of these models are expressed in terms of *differential equations*.

Importantly, all the fundamental laws of nature can be expressed in terms of differential equations.

Some famous differential equations:

► Newton's 2nd law of motion (particle): $\frac{d\mathbf{p}}{dt} = \mathbf{F}_{\text{net}}$

► Newton's 2nd law of motion (fluid continuum): $\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = \frac{1}{\rho} \nabla p + \mathbf{g} + \frac{1}{\rho} \nabla \cdot \mathbf{s}$

► Maxwell's equations (let there be light!):

$$\nabla \cdot \mathbf{E} = \rho/\epsilon, \quad \nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon \frac{\partial \mathbf{E}}{\partial t} \right)$$

► Schrödinger equation (foundation of all electronics): $i\hbar \frac{\partial}{\partial t} \psi(\vec{x}, t) = -\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{x}, t) + V(x, t) \psi(\vec{x}, t)$

Nobel Prize for Physics (1933).

► Black-Scholes equation (let there be money!): $\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$

Nobel Prize for Economics (1997).

Definition – Differential Equations

equations that contain the derivatives of one or more dependent variables with respect to one or more independent variables.

$$m \frac{d^2 x}{dt^2} = -kx, \quad \begin{array}{l} x \text{ dependent variable} \\ t \text{ independent variable} \end{array}$$

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(D(x) \frac{\partial u}{\partial x} \right), \quad \begin{array}{l} u \text{ dependent variable} \\ x, t \text{ independent variables} \end{array}$$

$$m \frac{d^2 x}{dt^2} + b \left(\frac{dx}{dt} \right)^2 + kx = A \sin \omega t, \quad \begin{array}{l} x \text{ dependent variable} \\ t \text{ independent variable} \end{array}$$

Ordinary Differential Equations (ODEs)

contain the ordinary derivatives of one or more dependent variables with just one independent variable

Example $m \frac{d^2 x}{dt^2} + b \left(\frac{dx}{dt} \right)^2 + kx = A \sin \omega t$

Partial Differential Equations (PDEs)

contain the partial derivatives of one or more dependent variables with two or more independent variables

Example $\frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(D(x) \frac{\partial u}{\partial x} \right)$

Definition – Order

the order of the highest derivative in the ODE.

Examples

$$\frac{dy}{dx} = \sin^2 y + \sin x \quad \text{order:}$$

$$\frac{d^2 y}{dt^2} + (\cos t) \frac{dy}{dt} = y \quad \text{order:}$$

$$\frac{d^2 y}{dx^2} \left(\frac{dy}{dx} \right)^7 - \frac{d^3 y}{dx^3} = e^x \sin x \quad \text{order:}$$

$$y'' + 2(y')^3 y - 3y^4 = e^y \quad \text{order:}$$

Definition – Analytic Solutions

functions which when substituted into the differential equation reduce the equation to an identity $\text{LHS} = \text{RHS}$

Example $m \frac{d^2 x}{dt^2} = -kx$ has the solution $x = \sin \sqrt{\frac{k}{m}} t$

Direct Integration

$$\frac{dy}{dx} = h(x)$$

Consider $y = \int_a^x h(s) ds$ where $h(s)$ is cts. on $s \in [a, x]$

From the Fundamental Theorem of Calculus

$$\frac{dy}{dx} = \frac{d}{dx} \left(\int_a^x h(s) ds \right) = h(x)$$

Hence $y = \int_a^x h(s) ds$ is the solution of $\frac{dy}{dx} = h(x)$

Note $y = \int_a^x h(s) ds + C$ is also a solution for any C

Definition – General Solution

the set of all possible solutions – often represented as a single equation containing arbitrary constants

Example $\frac{d^2y}{dx^2} = \sin x$

(3.2) Initial Value Problem (IVP)

Definition

an n th order ODE together with specified values for the solution and up to $n - 1$ derivatives at a given point.

Example

$$\frac{d^2y}{dx^2} = \sin x, \quad y(0) = 0, \quad y'(0) = 0$$

Boundary Value Problem (BVP)

Definition

an n th order ODE together with specified values for the solution and/or its derivatives (up to $n - 1$ derivatives) at more than one point.

Example

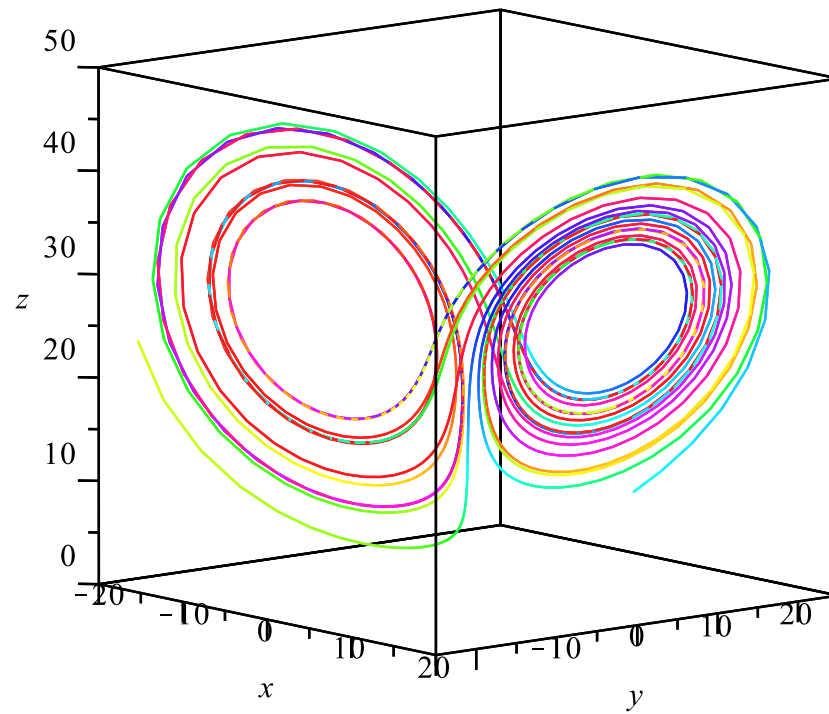
$$\frac{d^2y}{dx^2} = \sin x, \quad y(0) = 0, \quad y(\pi) = -\pi$$

Existence and Uniqueness of Solutions

- ▶ does a solution always exist?
- ▶ if a solution does exist
does it exist for all values of the independent variables?
- ▶ is there always a unique solution if a solution exists?
- ▶ how do we find solutions?

Analytic solutions may not exist but can be found numerically:

Lorenz Attractor



$$\begin{aligned}\frac{dx}{dt} &= -\sigma x + \sigma y \\ \frac{dy}{dt} &= -xz + rx - y \\ \frac{dz}{dt} &= xy - bz\end{aligned}$$

(3.3) First Order Separable ODEs

$$\frac{dy}{dx} = \frac{g(x)}{h(y)}$$

Multiply by $h(y)$ and integrate w.r.t. x then

$$\int h(y) \frac{dy}{dx} dx = \int g(x) dx \quad \Rightarrow \quad \int h(y) dy = \int g(x) dx$$

Notes:

- i) We replaced the derivative $\frac{dy}{dx}$ by a quotient (Leibniz notation) in the above.
- ii) Carry out the integrations if possible to find a function of y on the left and a function of x on the right
- iii) Solve to find an explicit expression for y as a function of x
- iv) If we have the IVP defined with $y(x_0) = y_0$ then we can use this as limits in the separated integrals

$$\int_{y_0}^y h(y) dy = \int_{x_0}^x g(x) dx$$

Example

$$\frac{dy}{dx} = y^2(1 + x^2), \quad y(0) = 1$$

Example

Solve $y \cdot (y')^3 = a$, $y(\xi) = \nu$, where a, ξ, ν are constants.

Example

Solve $\frac{dy}{dx} = \frac{y}{x(x-1)}$, for $y > 0$, $x > 1$

(3.4) First Order Linear ODEs

An n th order linear ODE can be written as

$$\frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + a_{n-2}(x) \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x)y = f(x)$$

- ▶ If $f(x) \equiv 0$ the ODE is homogeneous as well as linear
- ▶ If $a_0, a_1, \dots$ are constants (independent of x) then a general solution is possible – see Section (3.8).
- ▶ 2nd order linear ODEs with non-constant a_0, a_1, a_2 are studied in MATH2120
- ▶ Linear first order ODEs can always be solved by using an integrating factor

1st Order Linear – Integrating Factor

Write the 1st order linear equation in standard form

$$\frac{dy}{dx} + a(x)y(x) = f(x)$$

Suppose we try to integrate both sides w.r.t. x

$$\int \left(\frac{dy}{dx} + a(x)y(x) \right) dx = \int f(x) dx$$

We could simplify the LHS if the integrand was the derivative of a product. It is not for general $a(x)$.

Integrating Factor ctd.

- ▶ But suppose we multiply by some $\phi(x)$ first and then integrate

$$\int \left(\phi(x) \frac{dy}{dx} + \phi(x) a(x) y(x) \right) dx = \int \phi(x) f(x) dx$$

- ▶ Can we find a $\phi(x)$ for every $a(x)$ so that the integrand on the LHS is the derivative of a product?
- ▶ We can if $\phi(x)$ satisfies $\phi(x) a(x) = \frac{d\phi}{dx}$.
- ▶ The appropriate integrating factor ϕ is the solution of the separable ODE $\frac{d\phi}{dx} = \phi(x) a(x)$.
- ▶ The **integrating factor** is

$$\phi(x) = e^{\int^x a(x) dx}$$

(the upper limit x on the integral means ignore the constant of integration in the integral)

Integrating factor method for a first order linear ODE

$$\frac{dy}{dx} + a(x)y(x) = f(x)$$

Calculate the integrating factor

$$\phi(x) = e^{\int^x a(x) dx}$$

Multiply the 1st order ODE by the integrating factor

$$\begin{aligned}\phi(x) \frac{dy}{dx} + \phi(x)a(x)y(x) &= \phi(x)f(x) \\ \Rightarrow \frac{d}{dx} (\phi(x)y(x)) &= \phi(x)f(x) \\ \Rightarrow (\phi(x)y(x)) &= \int^x \phi(x)f(x) dx + C\end{aligned}$$

Finally

$$\Rightarrow y(x) = \frac{1}{\phi(x)} \int^x \phi(x)f(x) dx + C \frac{1}{\phi(x)}$$

Example

$$\frac{dy}{dx} - 2y = e^{3x}$$

Example

$$t^2 \frac{dy}{dt} + 2 t y = \sin(t)$$

Example

$$(x-1)^3 \frac{dy}{dx} + 4(x-1)^2 y = x+1, \quad \text{for } x-1 > 0$$

(3.5) First Order Exact ODEs

The ODE

$$F(x, y) + G(x, y) \frac{dy}{dx} = 0 \quad \text{or} \quad F(x, y)dx + G(x, y)dy = 0$$

is said to be exact if

$$\frac{\partial F}{\partial y} = \frac{\partial G}{\partial x}$$

If the ODE is exact then the solution $y = y(x)$ is defined implicitly by $H(x, y) = C$ (a constant) where

$$\frac{\partial H}{\partial x} = F \quad \frac{\partial H}{\partial y} = G$$

We know F and G but we need to solve two coupled PDEs to find H
But why does this work?

Let's show ODE $H(x, y) = C$ is the solution of an exact ODE

$$H(x, y) = C \quad \implies \quad dH = 0$$

We also know that changes dH due to changes dx and dy obey

$$dH = \frac{\partial H}{\partial x} dx + \frac{\partial H}{\partial y} dy$$

Thus, with $F(x, y) = \frac{\partial H}{\partial x}$ and $G(x, y) = \frac{\partial H}{\partial y}$

$H(x, y) = C$ is a solution of

$$dH = F(x, y) dx + G(x, y) dy = 0$$

and

$$\frac{\partial F}{\partial y} = \frac{\partial^2 H}{\partial y \partial x} = \frac{\partial^2 H}{\partial x \partial y} = \frac{\partial G}{\partial x}$$

Notes:

- ▶ The solution $H(x, y) = C$ defines y implicitly as a function of x . It is not always possible to get an explicit expression $y(x) = \dots$
- ▶ The following are equivalent

$$F(x, y) + G(x, y) \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{F(x, y)}{G(x, y)}$$

$$F(x, y) dx + G(x, y) dy = 0$$

- ▶ It is easy to check if the ODE is exact. We require $\frac{\partial F}{\partial y} = \frac{\partial G}{\partial x}$
- ▶ But how do we find the solution $H(x, y) = C$ from the coupled PDEs $F = \frac{\partial H}{\partial x}$ and $G = \frac{\partial H}{\partial y}$?

Constructing the solution

- ▶ Given $F(x, y) = \frac{\partial H}{\partial x}$ integrate w.r.t. x then

$$H(x, y) = \int^x F(x, y) dx + K(y) = f(x, y) + K(y)$$

where $f(x, y)$ is now known.

Now exploit fact that we also know $\frac{\partial H}{\partial y} = G(x, y)$

- ▶ Differentiate w.r.t. y then $\frac{\partial H}{\partial y} = \frac{\partial f(x, y)}{\partial y} + \frac{dK}{dy}$
- ▶ Equate $\frac{\partial H}{\partial y} = G \Rightarrow \frac{dK(y)}{dy} = G(x, y) - \frac{\partial f}{\partial y} = J(y)$, a function of y only!
- ▶ Now just integrate w.r.t. y to obtain $K(y) = \int J(y) dy$
- ▶ and build the solution $H(x, y) = f(x, y) + K(y) = C$ where $f(x, y)$ and $K(y)$ are now known functions.

Example

$$(2x + 3y) dx + (3x + 4y) dy = 0$$

Show this ODE is exact and solve.

Example

$$2x \sin(x^2 + y^2) + (2y \sin(x^2 + y^2) + 2y) \frac{dy}{dx} = 0$$

Solve: